

Telecom Revolution: 4G and Beyond

4G offers benefits but not without its share of problems. Nevertheless, the telecom world, ably supported by chipmakers, is all set to stride ahead into new 4G horizons

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This year, Bharti Airtel launched a '4G' service in Kolkata and Bengaluru, making India one of the first nations to adopt the newest baby in the wireless world. 4G or IMT-Advanced is the fourth generation in telecommunications, characterised by reliable lightning-speed broadband wireless access (BWA)—a dream come true for not just the young and raging population but also businesses, media agencies, governments and many more stakeholders. It is hoped that 4G will catalyse national development, as it means wireless communications to the nooks and cor-

ners, even across rugged terrains and the poorest of villages.

With the launch of Bharti Airtel's service, umpteen questions have arisen in the minds of consumers across the country—what really is 4G, what sets it apart as a new generation, is Bharti Airtel's service (based on the TDD-LTE standard) really 4G, what are the competing 4G standards, how is 4G doing worldwide, what are the pros and cons of this generation, and what is next in the pecking order? Here we try to answer these questions.

The fourth generation has just begun

Some service providers in countries like South Korea and Scandinavia

started terming their service as 4G as early as 2006. Others started doing so in the US a few years later. Humbug and marketing tricks!

4G is a very nascent technology and there are very few deployments worldwide that qualify as 4G today—even by a very liberal definition. If we keep to the true definition of IMT-Advanced, 4G is not even here yet.

"In the mid 2000s, as 3G systems were being deployed, there was the initial definition of 4G as providing 100Mbps bandwidth. Subsequent to 2006, many systems which exceeded 40-50 Mbps started using '4G' as a marketing ploy for their networks and some providers called their advanced 3G networks as '4G Lite' with 'Lite'



Customers get hands-on experience with Verizon Wireless flexible digital showcase of 4G